31 Beyond Default Topologies: Optimising the Meshroom Pipeline for High-Fidelity 3D Point Cloud Reconstruction and Acquisition Protocol Redesign

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This study addresses the inherent limitations of the default Meshroom photogrammetry pipeline, proposing an optimised methodology for high-fidelity 3D point cloud reconstruction and detailing its subsequent deterministic effects on the acquisition workflow. Standard automated processing often introduces topological artifacts due to down-scaling, closed-object assumptions inherent in default meshing algorithms (e.g., via Poisson surface reconstruction), and suboptimal feature matching across homogeneous surfaces or under non-calibrated optical settings. To overcome these constraints, we introduce a rigorously structured acquisition and processing protocol designed for high-resolution volumetric fidelity.Our methodology intervenes systematically at both the physical data-capture and computational pipeline levels. First, the acquisition phase dictates strict environmental control parameters, ensuring a rigid geometric image overlap matrix (>80%) and the correction of camera-related distortion. Second, by expanding the processing architecture beyond the baseline Meshroom environment, we systematically reconfigure the default statistical calculation of normals and mesh reconstruction. This includes the substitution of default feature extractors with a multi-algorithmic tensor (combining SIFT and AKAZE), bypassing automated scaling protocols in the DepthMap nodes, implementing a spatial filtering mechanism that preserves sharp geometric transitions without enforcing artificial surface closures, and integrating color correction procedures through a separate dataset within a fully Free Libre and Open Source pipeline.